

PATJ deficiency leads to cystic kidney disease and related ciliopathies

Daniel Epting,^{1,6,*} Daniela A. Braun,² Eva Decker,³ Elisabeth Ott,¹ Tobias Eisenberger,³ Nadine Bachmann,³ Pavel Nedvetsky,⁴ Michael P. Krahn,⁴ Friedhelm Hildebrandt,⁵ and Carsten Bergmann^{1,3,*}

Summary

Cystic kidney disease and related ciliopathies are caused by pathogenic variants in genes that commonly result in ciliary dysfunction. For a substantial number of individuals affected by those cilia-related diseases, the causative gene remains unknown. Using massively parallel sequencing, we here identified a pathogenic bi-allelic variant in the gene encoding PALS1-associated tight junction protein (PATJ) also known as inactivation-no-afterpotential D-like, INADL in an individual with ciliopathy. The affected fetus carried the homozygous truncating *PATJ* nonsense variant c.830delC (p.Pro277fsX), and presented with a syndromic phenotype mainly characterized by polycystic kidney disease and hydrocephalus. Using zebrafish (*Danio rerio*) as a vertebrate *in vivo* model organism, we could validate our patient findings and demonstrated a ciliopathy phenotype. In addition, we were able to address a hitherto not described role of Patj for cilia formation and function. Taken together, with the Crumbs cell polarity complex member PATJ, we add a new member to the large family of ciliopathy-related human disease proteins that is different from the classical ciliopathy protein classes, and may offer new perspectives for drug development.

Introduction

Virtually all non-dividing vertebrate cells possess either non-motile primary cilia or motile cilia. These protrude from the outside of the cells, acting respectively as mechano-sensors or creating a fluid flow. Reports of the last two decades have clearly shown that cilia-related defects are responsible for a huge and still growing number of genotypically and phenotypically variable human disorders, collectively termed ciliopathies. Among these are autosomal dominant and recessive polycystic kidney disease (PKD) (ADPKD/ARPKD; MIM: PS173900), nephronophthisis (NPH) (MIM: PS256100), and a number of syndromic phenotypes such as Joubert syndrome (MIM: PS213300), Meckel syndrome (MIM: PS249000), and Bardet-Biedl syndrome (MIM: PS209900).¹ Many causative genes are pleiotropic, resulting in an extensive list of potential clinical features such as cystic kidney and liver disease, congenital hepatic fibrosis, obesity, intellectual disability, retinal degeneration, polydactyly, hydrocephalus, infertility, and situs inversus. Despite remarkable progress in the identification of disease-causing genes and their encoded proteins related to cystic kidney disease or related ciliopathies, the genetic cause for a significant number of ciliopathy-affected individuals is still unknown. Emerging evidence indicates that proteins traditionally associated with epithelial polarity also play an important role in the etiology of ciliopathies. Some

members of the polarity complexes (Par and Crumbs complex) have, besides their well-described roles in apical-basal polarity and establishing/maintenance of cell-cell junctions, additional functions in cilia formation and function. Hence, Par complex members PAR3, PAR6, and aPKC co-localize at the primary cilium as well as the Crumbs complex member CRB3 (both isoforms CRB3A and CRB3B) and are required for proper cilia formation and function. Moreover, there is evidence that CRB3A interacts with the Par complex, thereby acting upstream of it and recruiting Par complex members to the cilium (reviewed in Bazellieres et al.²). Functional *in vivo* studies revealed that depletion of Crumbs proteins resulted in cilia abnormalities and ciliopathy-related phenotypes in zebrafish.^{3,4} In addition, *Crb3* knockout mouse models demonstrated perinatal lethality and cilia-related phenotypes such as cystic kidneys and improper airway clearance.^{5,6} The Crumbs complex member PATJ physically interacts and co-localizes with the nephrocystins NPHP1 and NPHP4 (both proteins are linked to NPH) that is probably important for epithelial morphogenesis.⁷ In addition, a physical interaction of PATJ and polycystin-2 (PC2) (linked to ADPKD) was described to be crucial for the regulation of PC2 channel activity and thereby probably plays a role for ADPKD pathogenesis.⁸ Only recently, it has been shown that PATJ depletion in kidney tubular epithelial cells results in disturbed apical-basal polarity, lumen formation in 3D cyst cultures, tight junction

¹Department of Medicine IV, Faculty of Medicine, Medical Center-University of Freiburg, University of Freiburg, Freiburg, Germany; ²Department Molecular Nephrology, Internal Medicine D (MedD), University Hospital of Münster (UKM), 48149 Münster, Germany; ³Medizinische Genetik Mainz, Limbach Genetics, Mainz, Germany; ⁴Medical Cell Biology, Medical Clinic D, University Hospital of Münster, Albert-Schweitzer Campus 1-A14, 48149 Münster, Germany; ⁵Department of Pediatrics, Boston Children's Hospital, Harvard Medical School, Boston, MA 02115, USA

⁶Lead contact

*Correspondence: daniel.epting@uniklinik-freiburg.de (D.E.), carsten.bergmann@medgen-mainz.de (C.B.)
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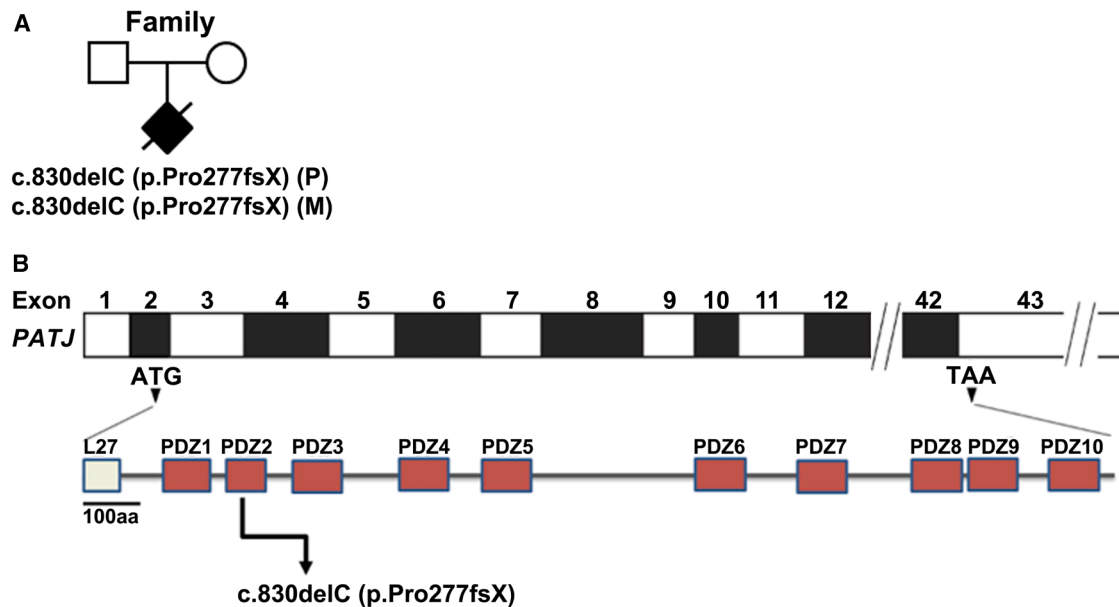


Figure 1. Identification of bi-allelic *PATJ* variants in an affected individual

(A) Pedigree for family with *PATJ* genetic variant c.830delC (p.Pro277fsX). Healthy parents and affected individual are shown in white and black boxes, respectively. *PATJ* genetic changes are shown below the symbol of the affected individual. The affected pregnancy was prenatally terminated (black symbol with transverse line); an autopsy was not performed.

(B) Schematic of human *PATJ* comprising 43 exons. Human *PATJ* encodes for a 1,801 amino acid protein with a single L27 domain and 10 PDZ domains (GenPept: NP_795352.3). The site of human *PATJ* variant identified is located in the second PDZ domain.

assembly, and cilia formation.⁹ Mechanistically, this study demonstrated that *PATJ* binds to and inhibits HDAC7 thereby regulating primary cilia formation. Notably, this function of *PATJ* does not require the interaction with *PALS1*, suggesting a novel role of *PATJ* in cilia formation, which is distinct from its known function in the Crumbs complex. However, an *in vivo* role of *PATJ* in ciliogenesis is still lacking, and no *PATJ* (MIM: 603199) variants have been described in patients so far. Here, we report the identification of a homozygous truncating *PATJ* nonsense variant in an individual with cystic kidney disease and related ciliopathies. In addition, our analyses in zebrafish demonstrate ciliopathy-associated phenotypes upon *Patj* depletion and a role of *Patj* in cilia formation and function.

Material and methods

Genetic analysis

Research was performed following written informed consent and according to the declaration of Helsinki. DNA extraction was performed according to standard procedures. NGS technologies and comprehensive bioinformatic analyses utilized in this project are described in detail elsewhere.^{10,11} Our approach is optimized in low-performance regions as well as in critical regions such as in *PKD1* as described.¹² High and reproducible coverage achieved by our sequencing approach also enabled copy-number variation (CNV) analysis. Performance of the wet-lab and bioinformatic processes are validated and controlled according to national and international guidelines reaching high sensitivity for SNV, Indels, and CNVs using well-established refer-

ence samples as well as a large cohort of positive controls, especially for CNVs.^{13,14} For interpretation of identified variants, we have developed our own algorithms using a stepwise filtering process conducted by an experienced team of scientists and supported by various bioinformatics decision tools. Sequence variants of interest were verified by Sanger sequencing if NGS results failed internal validation guidelines.

Zebrafish husbandry, lines and embryo maintenance

The fish used in this study were maintained at the Zebrafish Facility of the Medical Center of the University of Freiburg. All animal work, zebrafish maintenance, and staging of embryos has been conducted as described recently.¹⁵ The study was approved by the Institutional Animal Care of the Medical Center of the University of Freiburg and the Regional Council Freiburg (permit ID G-16/89). All methods were carried out in accordance with ARRIVE guidelines. The following wild-type (WT) and transgenic strains were used: AB/TL (WT), *li1Tg*,¹⁶ *cup^{tc321}*,¹⁷ and *elipsa^{tp49d}*.¹⁸

Results

We performed massively parallel sequencing of the exome of an affected fetus with cystic kidney disease and hydrocephalus originating from a consanguineous marriage for which we could not identify any pathogenic variant in known genes prior to this. These analyses resulted in the identification of an allegedly pathogenic *PATJ* nonsense variant c.830delC (p.Pro277fsX) in the homozygous state (Figure 1).

In order to better understand the function of *PATJ* and to validate our patient findings, we used the model

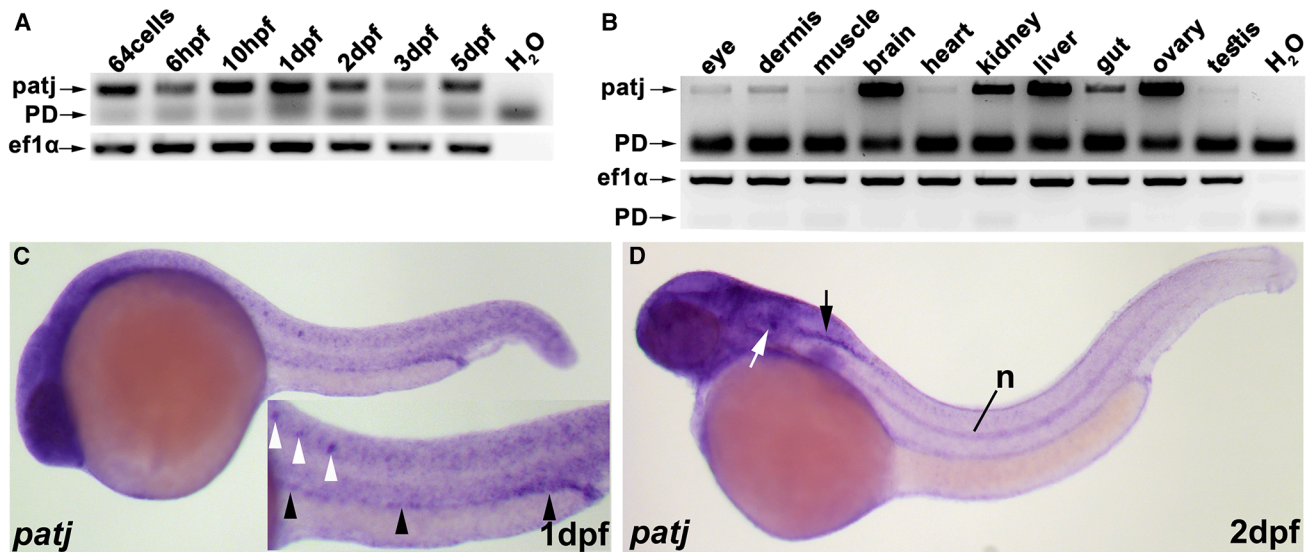


Figure 2. Expression analyses reveal specific expression of *patj* during embryogenesis and in adult organs in zebrafish
(A and B) Analyses of *patj* expression via semi-quantitative RT-PCR on cDNA of different embryonic developmental stages (A) or adult organs (B) in zebrafish, respectively. H₂O served as negative control and *ef1α* as loading control. PD, primer dimer.
(C and D) Analyses of *patj* expression via WISH at 1 and 2dpf; pronephric tubules (black arrowheads), neuronal cell populations in the spinal cord (white arrowheads), otic vesicle (white arrow), hindbrain (black arrow), and notochord (n). Embryos are shown from lateral with anterior to the left.

organism zebrafish, which offers unique advantages for analyzing various aspects of vertebrate development including ciliogenesis. We first analyzed the temporal and spatial expression of *patj* mRNA in zebrafish. To do this, we performed semi-quantitative reverse transcriptase-PCR (RT-PCR) on cDNA from different embryonic developmental stages and adult organs of zebrafish. These studies revealed that *patj* is expressed throughout early development and with different intensity levels in the analyzed organs with prominent expression in brain, kidney, liver, and ovary (Figures 2A and 2B). In addition, whole-mount *in situ* hybridization (WISH) analyses revealed expression of *patj* in highly ciliated tissues, e.g., pronephric tubules, spinal cord, and otic vesicles at 1 day post-fertilization (dpf) and 2 dpf (Figures 2C and 2D).

We also analyzed a potential ciliary role of Patj in zebrafish by using a morpholino (MO)-based knockdown approach. For this, we used two splicing-blocking MOs (*patj*-MO1 and *patj*-MO2), which, in zebrafish, respectively target intron4-exon5 and intron5-exon6 boundaries of *patj* pre-mRNA. The efficiency of MOs was validated by semi-quantitative RT-PCR, and showed, compared with the control, significantly reduced *patj* PCR-product levels in the Patj morphant embryos (Figure S1A). Knockdown with 4 ng of *patj*-MO1 or 6 ng of *patj*-MO2 resulted in significant hydrocephalus formation, otolith deposition defects, ventral body curvature, and a randomized heart looping (indicates defective left-right [LR] asymmetry) at 2 dpf, representing well-described ciliopathy-associated phenotypes in zebrafish (Figures 3A and S1B–S1G).¹⁵ Additionally, we performed

WISH analyses using the marker *southpaw* (*spaw*) as the earliest known LR asymmetry marker and *forkhead box protein a3* (*foxa3*) as a marker for primordial liver, pancreas, and intestine in zebrafish embryogenesis. These analyses clearly demonstrated defective LR asymmetry in Patj-deficient zebrafish embryos compared with control (Figures 3B and 3C). In zebrafish, the organ of laterality is known as the Kupffer's vesicle, containing cells with a single motile cilium. Several studies have shown that defective cilia in this organ ultimately lead to LR asymmetry defects.¹⁹ Analyses of cilia formation in the Kupffer's vesicle at the stage of eight somites showed that the cilia in the Patj morphants were significantly reduced in length compared with control embryos (Figure 3D). Noteworthy, motile cilia formation in the pronephric tubules (containing single ciliated and multiciliated cells) of Patj morphants appeared unaffected compared with control embryos at 1 and 2 dpf (Figure S2).

To verify the results of our MO experiments, we generated a Patj knockout in zebrafish (deletion of 1 bp in exon5 of *patj* leading to a frameshift and a premature stop codon) using the CRISPR-Cas9 technology (Figure S3). Surprisingly, maternal-zygotic (MZ) *patj* mutants did not display all of the ciliopathy-associated phenotypes that we observed in the Patj morphants, i.e., hydrocephalus formation and ventral body curvature; however, we detected striking LR asymmetry defects, analyzed by heart looping and by WISH analyses with the markers *spaw* and *foxa3* (Figures 4A, S4A, and S4B). Subsequent analyses of cilia in the Kupffer's vesicle revealed, compared with controls, significant shorter cilia in the MZ*patj* mutant embryos (Figure 4A). Multi-PDZ

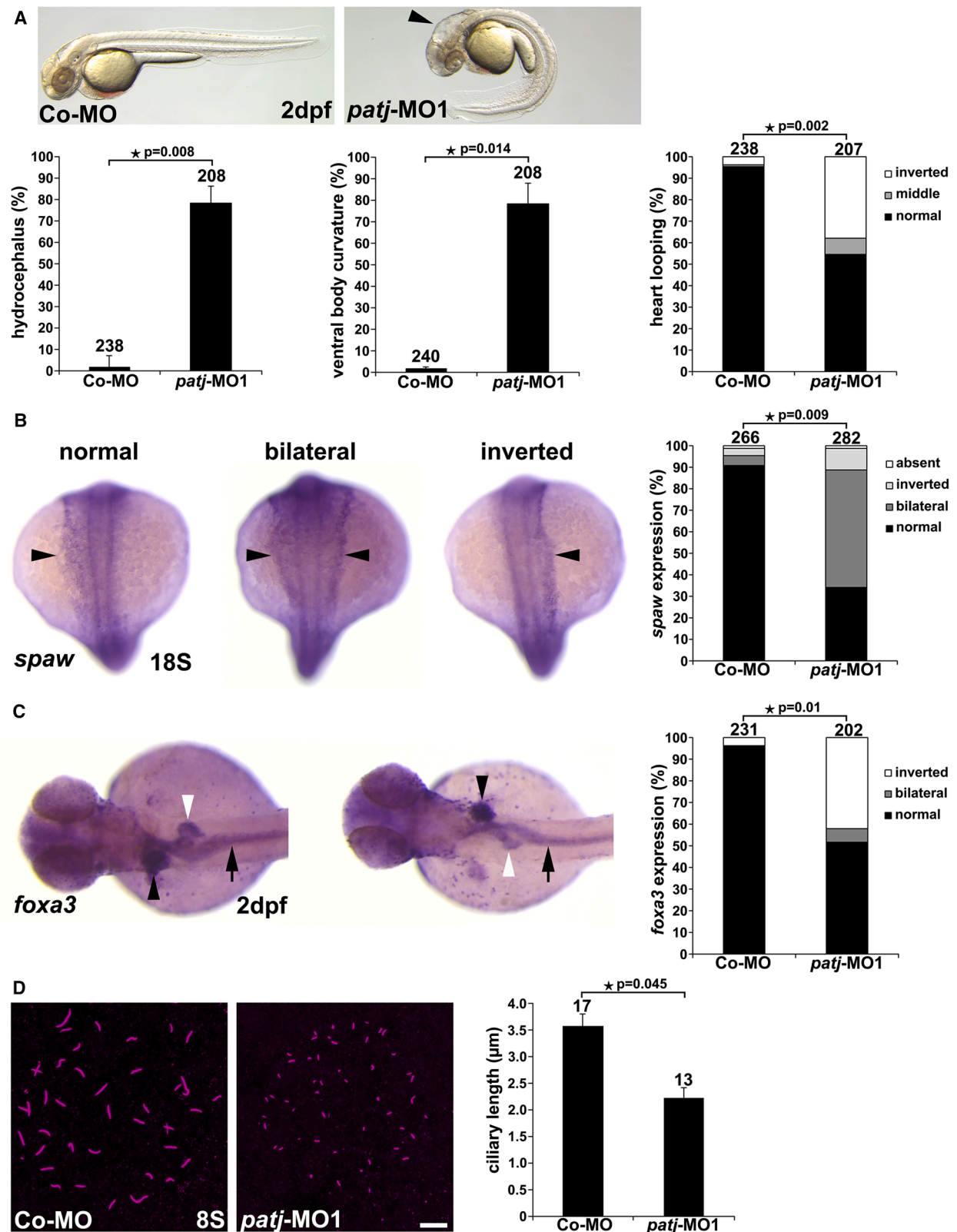


Figure 3. Knockdown of Patj results in ciliopathy-associated phenotypes in zebrafish

(A) Representative bright-field images of Co-MO- and *patj*-MO1-injected embryos at 2dpf; hydrocephalus (black arrowhead). Embryos are shown from lateral with anterior to the left. Quantification of hydrocephalus formation, ventral body curvature, and altered heart looping (analyzed as normal, middle (unlooped), and inverted) of Co-MO- and *patj*-MO1-injected embryos at 2dpf.

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domain protein 1 (MUPP1) represents a homolog of *PATJ* and, due to their high similarity, both proteins might act in a redundant fashion. Since we only observed ciliopathy-associated defects in early development of *MZpatj* mutant embryos, we considered the possibility that *Mpdz* (the zebrafish ortholog of MUPP1) might compensate for the loss of *Patj* later in development. Therefore, we generated a knockout for *mpdz* in zebrafish via CRISPR-Cas9 (deletion of 7 bp in *mpdz* leading to a frame-shift and a premature stop codon), and performed an in-cross of homozygous *mpdz* and *MZpatj* animals to generate double heterozygous *patj;mpdz* knockouts. In-crosses of animals with double heterozygous *patj;mpdz* knockout resulted in embryos with only a pericardial edema and others that show a dorsal body curvature in addition to the pericardial edema. Subsequent genotyping revealed, in accordance with Mendelian ratios, that the pericardial edema phenotype corresponds to homozygous deletion of *Mpdz*, whereas the phenotype with dorsal body curvature and pericardial edema results from the homozygous deletion of both proteins, *Patj* and *Mpdz* (Figure 4B). Cilia formation in the pronephric tubules of homozygous *mpdz* knockouts and double homozygous *patj;mpdz* knockouts appeared unaffected at 2 dpf (Figure S4C).

Recent studies in zebrafish revealed that proper body axis morphogenesis relies on cilia-dependent formation of the Reissner fiber (RF), an acellular and filamentous structure that is present in the cerebrospinal fluid.^{20,21} We therefore analyzed RF formation in double homozygous *patj;mpdz* knockouts by whole-mount co-immunostaining using antibodies for acetylated tubulin and RF. As controls, we included the well-described cilia-defective mutants *elipsa^{tp49d}* and *cup^{tc321}* that display ventral and dorsal body curvature, respectively.^{17,18} Defective RF formation has been described for *elipsa^{tp49d}*, but respective results have not been reported for the *cup^{tc321}* so far.²⁰ Our results revealed RF disorganization in *elipsa^{tp49d}*, *cup^{tc321}*, and double homozygous *patj;mpdz* knockout embryos compared with homozygous *mpdz* knockout and control sibling embryos at 2 dpf (Figures 4C and S4D). Moreover, we performed quantitative RT-PCR to analyze cilia-dependent signaling pathways in double homozygous *patj;mpdz* knockouts. While

the Wnt signaling pathway was unaffected, we observed, compared with the control, significant dysregulation of Hedgehog (Hh) signaling pathway components in double homozygous *patj;mpdz* knockout embryos compared with the control (Figure 4D).

Discussion

Our results conclusively identified *PATJ* as a novel cystic kidney disease and ciliopathy-related candidate gene. We report here a convincing bi-allelic nonsense *PATJ* variant in an individual affected by cystic kidney disease and related ciliopathies. The mutation results in a truncated version of the protein, most likely leading to a non-functional or degraded protein, which most probably explains the severity of the clinical manifestations. For the *PATJ* homolog MUPP1 (also known as MPDZ), pathogenic variants have been reported in affected individuals (from unrelated families) presenting with autosomal recessive non-syndromic congenital hydrocephalus.^{22,23} In addition, mouse *Mpdz* knockout models lead to the formation of hydrocephalus.²⁴ Notably, this report revealed morphological intact cilia on ependymal cells but demonstrated that MUPP1/MPDZ is essential for maintaining ependymal integrity. Another report has shown that loss of MPDZ in mouse results in hyperpermeability of the choroid plexus causing hydrocephalus formation.²⁵ To date, no *Patj* mouse knockout model has been reported. We have studied in the *in vivo* model organism zebrafish a hitherto undescribed role of *Patj* in cilia formation and function. RT-PCR and WISH analyses showed that *patj* is highly expressed throughout embryogenesis and in different ciliated embryonic and adult tissues. MO-mediated knockdown of *Patj* resulted in well-known ciliopathy-associated phenotypes, and analyses of cilia in the Kupffer's vesicle (LR organizer in zebrafish) revealed a significant reduction of ciliary length and thus most probably causes the observed LR asymmetry defects. A CRISPR-Cas9-mediated knockout of *Patj* recapitulates the observed *Patj* morphant phenotypes, showing reduced ciliary length in the Kupffer's vesicle and LR asymmetry defects, but no other ciliopathy-associated phenotypes could be documented.

(B) Quantification of WISH-analyzed Co-MO- and *patj*-MO1-injected embryos at 18 somites (S) using *southpaw* (*spaw*) as LR asymmetry marker. *Spaw* expression was analyzed in respect to its localization in the embryo as normal (on the left side, black arrowhead), bilateral (on both sides, black arrowheads), inverted (on the right side, black arrowhead), and absent (no expression). Embryos are shown from dorsal with anterior to the top.

(C) Quantification of WISH-analyzed Co-MO- and *patj*-MO1-injected embryos at 2dpf using *foxa3* as LR asymmetry marker. *Foxa3* expression was analyzed in respect to its localization in the embryo as normal (liver [black arrowhead] on the left side, pancreas [white arrowhead] on the right side and normal looping of the intestine [black arrow]), bilateral (liver, pancreas, and intestine in the middle of embryonic axis), and inverted (liver [black arrowhead] on the right side, pancreas [white arrowhead] on the left side, and reversed looping of the intestine [black arrow]). Embryos are shown from dorsal with anterior to the left.

(D) Representative confocal images of the Kupffer's vesicle of Co-MO- and *patj*-MO1-injected embryos at the stage of 8S immunostained with anti-acetylated tubulin as a ciliary marker. Scale bar, 10 μ m. Quantification of the ciliary length in the Kupffer's vesicle of Co-MO- and *patj*-MO1-injected embryos at 8S.

Number of embryos used for analyses are shown above respective bar. Data were analyzed by Student's *t* test (2-sided, unpaired); error bars represent the standard error of the mean (SEM). A *p* value of <0.05 was considered statistically significant.

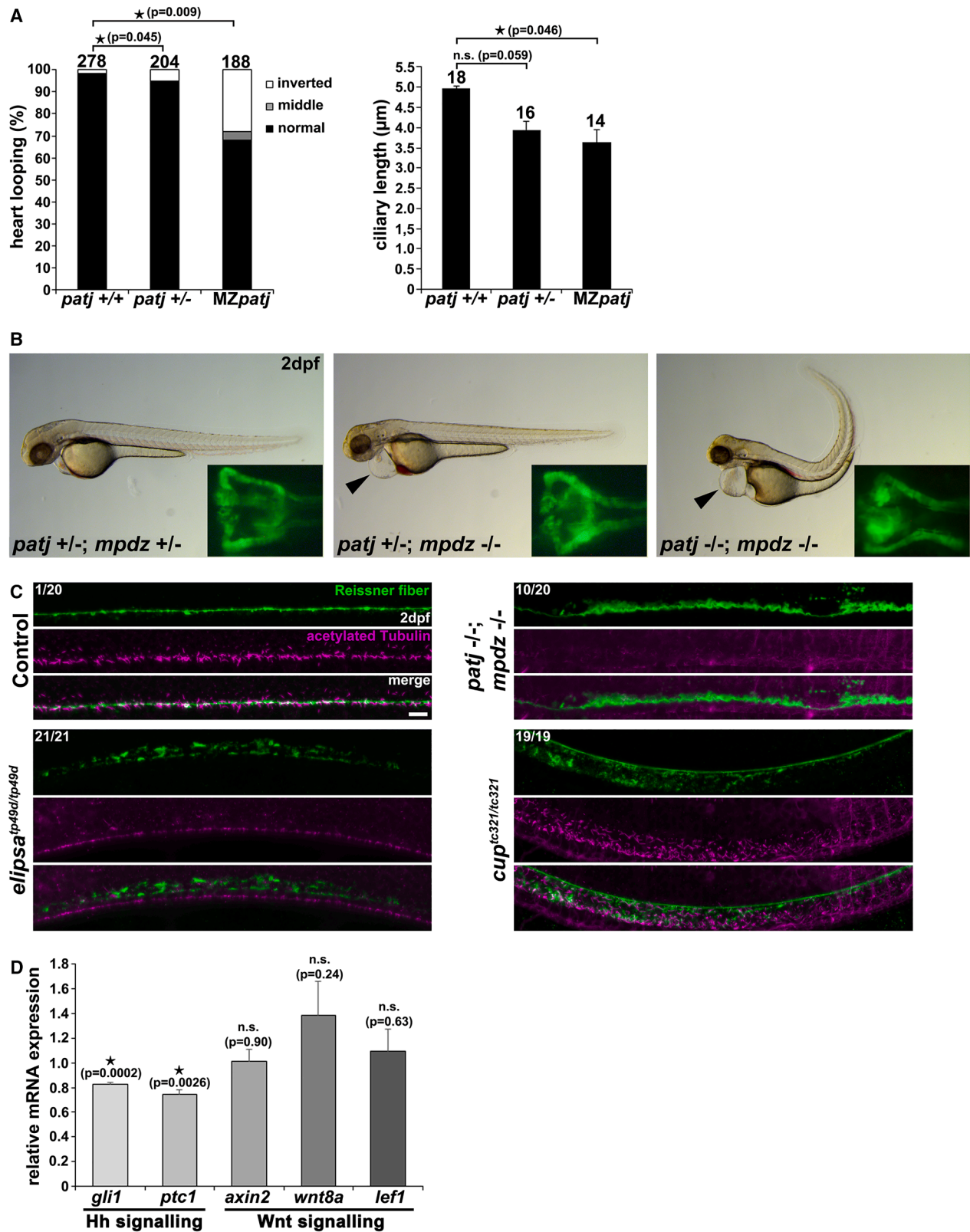


Figure 4. CRISPR-Cas9-induced *patj* and *patj;mpdz* zebrafish mutants display ciliopathy-associated phenotypes
 (A) Quantification of altered heart looping (analyzed as normal, middle (unlooped) and inverted), and of the ciliary length in the Kupfer's vesicle of wild-type (*patj* +/+), heterozygous *patj* mutants (*patj* +/-), and MZ*patj* mutants at 2dpf and 8S, respectively. Number of embryos used for analyses are shown above respective bars. Data were analyzed by Student's *t*-test (2-sided, unpaired); error bars represent the SEM. A *p* value of <0.05 was considered statistically significant.

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Mechanisms of genetic compensatory response have been well studied in zebrafish, and thus it is reasonable that the *Patj* homolog *Mpdz* compensates for the loss of *Patj* in later development of *Patj* knockout embryos.^{26,27} Notably, it has been reported that loss of *MPDZ* leads to increased expression of *PATJ* protein, and *in vitro* silencing of *MPDZ* resulted in increased *PATJ* mRNA expression.^{24,28} A double homozygous *patj;mpdz* knockout displayed pericardial edema (due to homozygous deletion of *Mpdz*) and a dorsal body curvature in zebrafish. Hence, we observed phenotypic discrepancy regarding body curvature between *Patj* morphants and *Patj;Mpdz* mutants, presenting with ventral and dorsal body curvature, respectively. Possibly the *Patj;Mpdz* double knockout results in a yet unknown genetic compensation mechanism that might have an influence in body axis formation and probably results in this context in a dorsal curvature phenotype. Of note, compensation mechanisms and their investigation are likely to be complex and far from being completely understood.²⁹ Moreover, in *Patj* morphants and *Patj;Mpdz* mutants maternal *patj/mpdz* mRNA and/or *Patj/Mpdz* protein might also have an influence in the phenotypic variability that we observed in our knockdown and knockout approaches. The majority of reported ciliopathy-associated zebrafish morphants and mutants display a ventral body curvature, but also a dorsal body curvature was reported, e.g., *Pc2* and *Bicc1* loss-of-function.^{17,30} Our analyses revealed defects in the formation of the RF in *Pc2* and *Patj;Mpdz* double knockouts which are most probably a result of defective cilia formation and function in the floor plate. Future studies are needed to analyze whether *Patj* potentially interacts directly or indirectly with PKD disease proteins *Pc2* and/or *Bicc1*. Defective primary cilia function often results in the dysregulation of important signaling pathway components. Indeed, we identified dysregulated expression of cilia-associated Hh signaling pathway components, thus further supporting an essential ciliary role of *Patj*. Our results provide further insight into the poorly characterized function of polarity complex

members in cystic kidney disease and related ciliopathies, and might be useful for respective drug discovery and therapeutic approaches.

Data and code availability

Exome sequence data were generated during clinical testing; however, study individuals were not consented for data sharing. Other datasets used and/or analyzed during the current study are available from the corresponding authors on reasonable request.

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Declaration of interests

The authors declare no competing interests.

Supplemental information

Supplemental information can be found online at <https://doi.org/10.1016/j.xhgg.2025.100514>.

(B) Representative bright-field images of a homozygous *mpdz* mutant (*patj* +/–; *mpdz* –/–) displaying pericardial edema (black arrowhead) and of a double homozygous *patj;mpdz* mutant (*patj* –/–; *mpdz* –/–) displaying pericardial edema (black arrowhead) and dorsal body curvature compared with a respective control clutch embryo (*patj* +/–; *mpdz* +/–) at 2dpf. Quantitative analysis from randomly selected control clutch embryos (without phenotype) revealed following genotypes: 3x (*patj* +/+; *mpdz* +/+), 3x (*patj* +/+; *mpdz* +/–), 5x (*patj* +/–; *mpdz* +/+), 5x (*patj* +/–; *mpdz* +/–), 2x (*patj* –/–; *mpdz* +/+) and 6x (*patj* –/–; *mpdz* +/–). Quantitative analysis from randomly selected embryos displaying pericardial edema revealed following genotypes: 9x (*patj* +/+; *mpdz* –/–) and 15x (*patj* +/–; *mpdz* –/–). Quantitative analysis from randomly selected embryos displaying pericardial edema and dorsal body curvature revealed following genotypes: 24x (*patj* –/–; *mpdz* –/–). The respective inserts show a fluorescent image of EGFP expression of the same embryo (dorsal view) indicating no detectable pronephric cyst formation for all three different genotypes. Embryos are shown from lateral with anterior to the left.

(C) Representative confocal images of *cup*^{tc321/tc321}, *elipsa*^{tp49d/tp49d}, *patj* –/–; *mpdz* –/– mutant embryos and respective control sibling embryos at 2dpf immunostained with anti-RF and anti-acetylated tubulin as a ciliary marker. Numbers represent embryos displaying RF disorganization and embryos that have been analyzed in total. Scale bar, 10 μm.

(D) Quantitative RT-PCR analyses reveal unaltered expression of Wnt signaling components *axin2*, *wnt8a*, and *lef1* while Hh signaling components *gli1* and *ptc1* were significantly downregulated in double homozygous *patj;mpdz* mutant embryos compared with control sibling embryos at 2dpf ($E^{-\Delta\Delta CT}$, normalized to control samples for all genes). Data were analyzed with Graphpad Prism software and one sample *t* test; error bars represent the SEM.

Web resources

OMIM, <http://www.omim.org/>

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